

PREET PATEL

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Education

Master of Science in Computer Science | University of Illinois, Chicago **Expected: May 2027**

Coursework: Cloud Computing, Computer Systems Security, Secured Web-Application Development

Bachelor of Technology in Computer Science | Ahmedabad University **May 2025**

Coursework: Artificial Intelligence, Machine Learning, Advanced Business Analytics, Computer Networks

Internship Experience

yourEDUGATOR

May 2025 – August 2025

Product Development Intern

Ahmedabad, Onsite

- Designed and engineered the IELTS Speaking module using Azure Speech SDK, ElevenLabs STT, and OpenAI TTS, improving end-to-end speech evaluation throughput by 3×.
- Built ETL pipelines to ingest synthetic JSON-schema datasets across Reading/Listening/Writing, improving data quality and reducing manual processing time by 70%.
- Collaborated directly with CEO to develop an automated IELTS-Writing Task-1 diagram generator (OpenAI + DALL-E) used in production to expand question coverage by 120%.
- Drove product ideation, crafted investor pitch decks, and presented the product features to the clients.

Techsture Technologies Ind. Pvt. Ltd.

May 2024 – August 2024

Research Intern

Ahmedabad, Onsite

- Deployed real-time C-V2X-based Pedestrian Detection and Alert Dissemination ITS testbed; validated 10+ scenarios demonstrating the robustness; achieved <10ms latency with multi-threaded UDP + SNMP monitoring.
- Built a TCP/DNS-based alert pipeline between server-RSU-OBUs, improving vehicular alert reliability and reducing drop rates.
- Designed a JavaFX dashboard with OpenStreetMap to visualize live vehicle telemetry and system alerts.

Technical Skills

- Languages:** Python, JavaScript/TypeScript, Go, Scala, C/C++, Java
- Databases:** MySQL, Oracle, MongoDB, PostgreSQL
- Technologies:** PowerBI, AWS (EC2, EMR, S3, EKS, Lambda), Kubernetes, Git, Docker, LaTeX
- Frameworks:** Node.js, Next.js, React.js, Flask, Django, RESTful APIs, FastAPI, OpenCV, MindAR, LangChain
- Distributed Systems/Data:** Spark, Hadoop (HDFS/YARN), Flink, GraphRAG, Lucene, Kafka, Neo4j

Projects

Distributed Word Embeddings with Hadoop and AWS EMR

September 2025 – November 2025

- Engineered a distributed pipeline using Hadoop MapReduce + Lucene to ingest and index 600+ PDFs, building stable chunk/embedding stores with content-hash deduplication and efficient k-NN search over float-vector fields.
- Implemented an incremental, fault-tolerant layer using Spark + Delta Lake, enabling re-indexing when documents or embedding models changed, and reducing recomputation by 90%+ through deterministic delta detection and metadata.
- Designed a cloud-native GraphRAG system on AWS using Flink, Neo4j, and Ollama inference, streaming chunks through relation extraction, LLM scoring, and idempotent graph upserts inside a Flink-on-EKS deployment.

Campus ARt — AR - based Collaborative Campus Experience

October 2025 – November 2025

- Built a full-stack AR web platform with real-time geolocation tracking, dynamic hotspot creation, and WebSocket-based multi-user synchronization across the UIC campus.
- Engineered the AR subsystem using MindAR, implementing marker compilation, multi-marker recognition, and drag-and-drop placement of drawings/3D objects directly into the camera feed.
- Designed persistent shared canvases + 3D object placement with dual-environment storage (local FS + Vercel Blob) and SQLite/Postgres switching, supported by comprehensive API + DB documentation.

RAND-SLA: A Randomized Bayesian Structure Learning Algorithm

September 2024 – November 2024

- Introduced a novel randomized algorithm for efficient skeleton learning of Bayesian Networks.
- Achieved faster convergence with up to 34% less computation time by reducing redundancy in Conditional Independence tests while maintaining competitive accuracy on benchmark datasets.

Leadership & Activities

- Research Scholar, MICxN Lab:** Led 5+ workshops and reproduced 15+ research papers in the tenure of 1.5 years. Reviewed 50+ manuscripts of the junior research scholars.
- Clash of Codes:** Team ranked 1/20 in a competitive coding-based team competition organised by the university in season 1 (2023) and 2/20 in season 2 (2024); 4/80 in individual scores and 3rd in problems solved.
- Teaching Assistant & Peer Tutor, Ahmedabad University:** Assisted the professors in courses like AI, Embedded Systems, Computer Networks, and Predictive Analytics for Business (classes up to 130 students).